TEXORA · SESSIONAL NOTES · STUDY SHEET

Garments Inspection

Performing a full garment inspection cycle — in-line check, end-of-line 100% check, measurement audit and a final AQL audit to ISO 2859-1 — on a production lot, with defect classification, DHU calculation and a documented pass/fail decision.

AM-4102 · ADVANCED APPAREL MANUFACTURING-I SESSIONAL · STUDY 3 · LEVEL 4 TERM 1

1. Aim and place of the study

Aim: to inspect garments at each control stage of a real production lot, classify defects as critical, major or minor, compute the in-line DHU, and run a final acceptance audit by ISO 2859-1 single sampling — arriving at a defensible ship / hold decision with the paperwork to support it.

Inspection is the measurement half of quality control: Study 10 of TE-3107 built the theory; this study performs it. The governing ideas carry over unchanged — defects are judged per garment by their worst class, and the sampling chain is **lot size + inspection level → code letter → sample size → Ac/Re at the chosen AQL**. The reference case used throughout Texora is: **lot 2,400 → code K → n = 125 → majors (AQL 2.5) Ac 7 / Re 8; minors (AQL 4.0) Ac 10 / Re 11; criticals Ac 0**. This study's audit uses exactly that case, on a real lot of 2,400 washed jeans from Studies 1–2 production.

2. Materials, station and specimens

Inspection station: inspection table with standard lighting (D65 or buyer-specified), viewing distance about 1 m; calibrated measuring tape and the style's points-of-measurement (POM) chart; defect catalogue with photographs; sealed approval (PP) sample and wash standard; needle detector access; audit forms, tally sheets and a random-number method for sample selection (carton/polybag list).

Specimens and stages: (i) in-line: 250 garments at the end-of-line table before packing; (ii) measurement audit: 13 garments drawn across sizes; (iii) final audit: the statistical sample of 125 from the packed lot of 2,400.

3. The inspection stages and defect classes

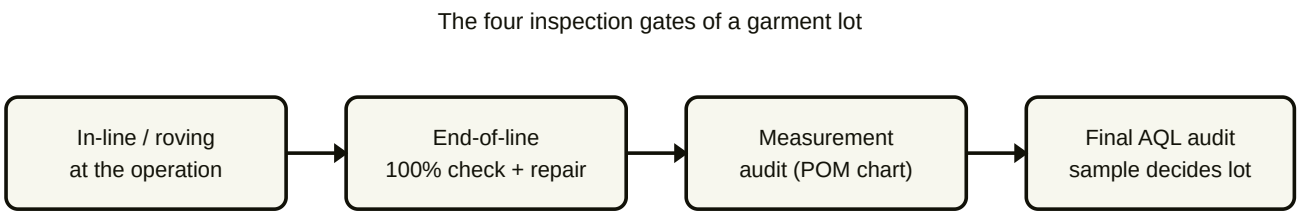


Figure 1 — Inspection gates. Each gate has its own method, sample basis and decision.

CLASS	RULE	EXAMPLES ON WASHED DENIM GARMENTS
Critical	Ac = 0 — one occurrence holds the lot.	Broken needle fragment (detector fail), wrong fibre-content label, mould or chemical smell (residual chlorine), sharp exposed pin.
Major	Counted at AQL 2.5.	Open seam, broken stitching at stress point, hole from stone damage, wrong size label, measurement beyond tolerance, heavy stain, non-functional button/zip.
Minor	Counted at AQL 4.0.	Untrimmed thread, slight crease mark, small slub, minor shading inside one garment, loose hanging thread at hem.

4. Worked calculations for this study

DHU AND THE AUDIT DECISION — VERIFY EVERY NUMBER

In-line gate: 250 garments checked, 18 defects recorded (a garment may carry more than one defect in DHU counting). DHU = (defects ÷ units inspected) × 100 = 18 ÷ 250 × 100 = **7.2**. Above the factory's end-of-line target of 5 — the line is told to find the dominant defect before the audit, not after.

Final audit (the reference case): lot 2,400, General Inspection Level II → code **K** → sample **n = 125**. Findings in the sample: **5 major-defective** garments and **9 minor-defective**, no criticals. Majors: 5 ≤ Ac 7 — pass. Minors: 9 ≤ Ac 10 — pass. Criticals: 0 — pass. **Decision: the lot is accepted for shipment**, and the report records all three counts, because each class is decided against its own AQL. (Compare TE-3107 Topic 10, where 12 minors failed the same lot size at Re 11 — the method is identical; only the findings differ.)

5. Procedure

1. **Prepare:** lay out the sealed sample, POM chart, defect catalogue and wash standard; check the tape against a steel rule; confirm lighting.
2. **In-line gate:** inspect the 250 garments systematically — front, back, inside, then measurements spot-checks — tallying every defect by type and class on the tally sheet. Compute DHU and the top-three defect Pareto on the spot.
3. **Repair loop:** send defectives to repair; repaired garments return for 100% re-check (never into the audit sample pool unexamined).
4. **Measurement audit:** measure the 13 drawn garments at every POM; record actual vs spec and tolerance; any POM systematically out of tolerance (same direction, several garments) is a lot fault, not a sampling matter.
5. **Final audit sampling:** list the packed cartons; select cartons and garments by the random method so the 125 come from the whole lot, all sizes and both packing days.
6. **Audit inspection:** inspect the 125 exactly as at the end-of-line gate, classifying each defective garment once, by its worst defect. Pass every garment over the needle detector (or check the detector log for the lot).
7. **Decide and document:** compare counts with Ac/Re per class; write the audit report (lot, code, n, findings per class, decision, inspector, date); a failing lot is held for 100% sort/repair and a fresh audit — the original sample is never "topped up" to reach Ac.

Observation points — what to record

- The defect Pareto from the in-line gate: defect type, count, class — and the operation responsible.
- DHU before and after the repair loop.
- Measurement audit table: any POM at or beyond tolerance, and whether deviations are random or systematic.
- Audit sampling integrity: how randomness was achieved; which cartons contributed.
- The final counts per class against Ac/Re, and the signed decision.

Precautions

- Inspectors rotate and take breaks: visual fatigue silently raises the miss rate after about an hour of continuous checking.
- Never inspect against memory — the sealed sample and catalogue are on the table, or judgement drifts within a shift.
- A critical defect stops the line, not just the lot: find the source (needle log, detector) before production continues.

- Sampling honesty: replacing a defective sample garment "because it is untypical" destroys the audit's meaning — and the buyer's auditor will find the same garments.

Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. An inspector checks 400 garments and records 26 defects. Calculate the DHU and state whether it meets an end-of-line target of 5.
2. In this study's audit, what would the decision be if the findings were 5 majors and 11 minors? Justify by Ac/Re.
3. Why is a garment with two majors and one minor counted once in the audit, and in which class?
4. The measurement audit finds waist readings 0.8 cm small on 11 of 13 garments, tolerance ±0.5 cm. Why is this not decided by the AQL counts?
5. Why must the audit sample be drawn randomly from packed cartons rather than taken from the top of the production line?

Answer key

1. DHU = 26 ÷ 400 × 100 = 6.5 — above target 5; the line must act on the dominant defect before audit.
2. Fail. Minors 11 ≥ Re 11 at AQL 4.0 for code K — one failing class fails the lot, even though majors pass (5 ≤ 7). The lot is held for 100% inspection/repair of the minor class and re-audited.
3. Sampling plans are calibrated on defective units. The garment counts once, as a major defective — its worst defect sets the class.
4. It is a systematic fault: the pattern/process is producing out-of-tolerance garments as a rule, so the lot fails on measurement regardless of the visual sample counts. Sampling decides random variation; a biased process must be corrected, not sampled around.
5. The audit judges the lot as shipped — all sizes, cartons and packing times. Line-top garments are a convenience sample, biased to one moment and one size run; defects concentrated elsewhere in the lot would be missed by design.
